

Question	Marking Guidance	Mark	Comments
09.1	Tissue and organ;	1 (1 x AO1)	Answers must be in this order
Question	Marking Guidance	Mark	Comments
09.2	(Q) starch (grain) OR lipid/fat (droplets) and (R) stroma;	1 (1 x AO1)	Reject stoma
Question	Marking Guidance	Mark	Comments
09.3	1. Fewer grana/lamella(e)/thylakoids; 2. (So) less chlorophyll and less light absorbed so less photosynthesis;	2 (2 x AO3)	Accept converse for healthy plants 1. Accept thinner Or smaller for fewer 2. Accept 'less light-dependent reactions' for 'less light absorbed so less photosynthesis' 2. Ignore LDR
Question	Marking Guidance	Mark	Comments
09.4	1. (Mutant) has less chlorophyll (but) more photosynthesis; 2. (Mutation) increased (chlorophyll) a:b OR (Mutation) increased (chlorophyll) ratio; 3. (Mutant) absorbs more light;	3 (3 x AO2)	2. Accept 'relatively more a than b' OR 'proportion of a to b increases' for ratio 3. Reject if more chlorophyll a in mutant 3. Accept (Mutant) absorbs more OR greater range of (light) wavelengths
